

# ESTEBAN SCHAFIR

Data Engineer · Applied AI Researcher · Ph.D. Candidate in Computer Science

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## SUMMARY

Data and AI engineer with 10+ years of combined experience in IT, data, and research: 4+ years of re-search at Florida International University and 7 years in enterprise IT and information security. I hold an M.S. in Cybersecurity and am pursuing a Ph.D. in Computer Science. I design and build end-to-end data pipelines in Python and SQL that collect, clean, transform, and integrate data from many sources, and I query relational and NoSQL databases (SQL Server, Oracle PL/SQL, PostgreSQL, MySQL, MongoDB). I also built an agentic Text-to-SQL engine that turns natural-language questions into complex SQL queries. I deliver data to analytics and machine learning models, KPI dashboards, and reports. My security background (ISO 27001, NIST, access management) means I treat data quality, reliability, and compliance as requirements from the start. I have also published on privacy and security challenges in e-Health.

## TECHNICAL SKILLS

|                                       |                                                                                                                                                                                      |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Engineering &amp; ETL/ELT</b> | Data pipelines, ETL/ELT, data integration, data transformation, data cleaning & normalization, data quality & validation, data modeling, database & schema design, data distribution |
| <b>Databases &amp; SQL</b>            | SQL, SQL Server, Oracle PL/SQL, PostgreSQL, MySQL, SQLite, MongoDB (NoSQL), vector databases (Chroma, Pinecone), complex query writing, Text-to-SQL                                  |
| <b>Programming &amp; APIs</b>         | Python (Pandas, NumPy), Java, C#, FastAPI, RESTful APIs & web services, JSON                                                                                                         |
| <b>Cloud &amp; DevOps</b>             | AWS (basic experience), Docker, Git, GitHub, GitLab                                                                                                                                  |
| <b>Analytics &amp; Visualization</b>  | Power BI (working knowledge), Excel (dashboards, pivot tables, charts), Matplotlib, KPI & metrics reporting, technical documentation                                                 |
| <b>Machine Learning &amp; AI</b>      | Scikit-learn, PyTorch, TensorFlow, Hugging Face, LLMs, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, predictive modeling, pattern discovery                            |
| <b>Data Security &amp; Compliance</b> | Data security, ISO 27001, NIST, risk assessment, auditing, Identity & Access Management (IAM), Privileged Access Management (PAM)                                                    |
| <b>Languages</b>                      | Spanish (native), English (fluent)                                                                                                                                                   |

## PROFESSIONAL EXPERIENCE

### Graduate Research & Teaching Assistant

Apr 2022 – Present

Florida International University

Miami, FL

- Built DecoSearch, an agentic Text-to-SQL engine that translates natural-language questions into complex SQL queries over relational databases, using complexity-aware query routing and plan-level repair to improve accuracy and efficiency.
- Designed and maintained end-to-end Python data pipelines (Pandas, NumPy) that collect, clean, transform, and augment large, complex datasets from different sources, preparing them for machine learning, statistical evaluation, and benchmarking.
- Achieved sub-10ms visual document search by distilling the multi-step reasoning of LLM-based retrievers into offline sparse indexes, turning a costly online process into a fast, scalable lookup.
- Developed and deployed machine learning models (PyTorch, Hugging Face, LangChain), including graph neural networks and agentic RAG systems, to find hidden patterns in data and automate tasks.
- Reported results in tables and figures (Matplotlib) across 7 peer-reviewed publications and preprints, and communicated findings to faculty and research collaborators.
- Supported courses as a Teaching Assistant, grading student assessments and tracking student performance data.

### Information Security Officer

Equifax Argentina SA

Sep 2017 – Nov 2021

Buenos Aires, Argentina

- Integrated and merged data from multiple sources (vulnerability scanners, audits, third-party vendor reviews) into consolidated datasets, cleaning and normalizing them for analysis.
- Maintained tracking databases of risks, findings, and remediation progress, so that issues could be followed reliably over time.

- Built recurring KPI dashboards, metrics reports, and figures in Excel on risk, vulnerability, and compliance status for leadership.
- Reduced critical infrastructure vulnerabilities by over 50% through remediation programs measured and tracked with these metrics.
- Directed enterprise risk assessments, security control implementations, and third-party vendor reviews, keeping systems and data compliant with security standards.

### Information Security Analyst

Puente Hnos

Mar 2016 – Sep 2017

Buenos Aires, Argentina

- Drove a 70% reduction in internal audit findings by tracking findings in a central database and reporting mitigation progress to management.
- Compiled and reported on data extracted from SIEM, ITSM, and access-management systems, using spreadsheets, dashboards, and summary reports.
- Implemented and managed SIEM, ITSM, and privileged access management tooling to protect enterprise systems and data from unauthorized access.

### Information Security Analyst

Cybsec by Deloitte

Oct 2014 – Mar 2016

Buenos Aires, Argentina

- Merged and enriched assessment outputs (Nessus, Nmap) into spreadsheets, and compiled them into client reports with tables and charts.
- Cut client audit findings by 60% through targeted compliance reviews and system hardening across diverse client environments.
- Worked closely with corporate clients to present results and design risk mitigation strategies.

## EDUCATION

### Ph.D. in Computer Science

Florida International University, Miami, FL

2024 – Present

GPA: 3.97/4.0

### M.S. in Cybersecurity

Florida International University, Miami, FL

2023

GPA: 4.0/4.0

### B.S. in Programming

Universidad Nacional de Quilmes, Buenos Aires, Argentina

2017

## PUBLICATIONS

- **Schafir, E.**; Zheng, X.; Salehi, H. A.; Chen, Z.; Sha, M.; Cheng, W.; Luo, D. (2026) DecoSearch: Complexity-Aware Routing and Plan-Level Repair for Text-to-SQL. *arXiv:2606.17821*.
- Jara, H.; **Schafir, E.** (2014) e-Health: An Introduction to the Challenges of Privacy and Security. *IEEE CONCAPAN XXXIV*. DOI: 10.1109/CONCAPAN.2014.7000407.
- Chen, Z.; Salehi, H. A.; **Schafir, E.**; et al. (2026) Addressing Structural Distribution Shift in Explanations for Graph Neural Networks. *IEEE TPAMI*. DOI: 10.1109/TPAMI.2026.3690304.
- Chen, Z.; Ni, J.; Salehi, H. A.; Zheng, X.; **Schafir, E.**; et al. (2026) Explanation-Preserving Augmentation for Semi-Supervised Graph Representation Learning. *AAAI*. DOI: 10.1609/aaai.v40i25.39183.
- Zheng, X.; Chen, Z.; **Schafir, E.**; et al. (2025) Towards Structurally Explainable Machine-Generated Text Detection: A Graph-Perspective Framework. *arXiv:2505.12507*.
- Ousat, B.; **Schafir, E.**; Kharraz, A. (2024) The Matter of Captchas: An Analysis of a Brittle Security Feature on the Modern Web. *ACM Web Conference*. DOI: 10.1145/3589334.3645619.
- Tofighi, M. A.; Ousat, B.; **Schafir, E.**; Kharraz, A. (2024) Constructs of Deceit: Exploring Nuances in Modern Social Engineering Attacks. *DIMVA*. DOI: 10.1007/9783031.6417186.